

Siddharth Korukonda

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EDUCATION

Stony Brook University

2023 – 2027

Bachelor of Science in Computer Science (Dean's List)

Stony Brook, NY

Minor in Applied Mathematics and Statistics, Specialization track in Artificial Intelligence and Machine Learning

Courses: Theory of Computation, Analysis of Algorithms, Data Structures, Object-Oriented Programming, Systems Fundamentals, Programming Abstractions, Linear Algebra, Calculus I-IV, Data Mining, Probability Theory, Game Theory, Graph Theory, Software Development, Machine Learning, Artificial Intelligence, Logic, Scripting

AWARDS AND CERTIFICATIONS

Mathematical Association of America (MAA): AIME - 5 (AMC 12B - 106.5)

Hackathons Won: Columbia DivHacks 2025, SBUHacks 2025, HopperHacks 2026

JP Morgan Chase: Quantitative Reserach Job Simulation, Software Engineer Job Simulation

Amazon: Certified Cloud Practitioner (in progress)

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, MySQL, C/C++, C#, Lua, JavaScript, HTML/CSS, Bash, TypeScript, LaTeX, R, MIPS

Frameworks: React.js, Node.js, Express.js, Three.js, .NET, Django, Spring Boot, MongoDB, PyTorch, Flask, Tailwind, REST APIs

Developer Tools: Postman, Git, LangChain, Hugging Face, PyCharm, IntelliJ, Kubernetes, Docker, AWS CLI, Jira, Pinecone, Snowflake

Libraries: NumPy, Matplotlib, Scikit, SciPy, Pydantic, OpenCV, TensorFlow, OpenPose, Asyncio, ONNX Runtime, Transformers, Pandas

Concepts: NLP, CI/CD, Computer Vision, 3D Gaussian Splatting, Software Security, Operating Systems, Cloud Computing, AWS, MCP Servers, Time Series Analysis, Regression Analysis, Statistic Analysis, Market Modeling

EXPERIENCE

Software Developer Co-op (Incoming)

August 2026 – December, 2026

IBM

San Jose, CA

Undergraduate Teaching Assistant

January 2026 - Present

Stony Brook University

Stony Brook, NY

- Created **assignments and labs** for **CSE 220 (System Fundamentals I)**.
- Delivered assistance to students in **lectures of approximately 200 students**.
- Graded assignments and held **weekly recitations and office hours**.

Software Developer

January 2025 – Present

Stony Brook University Hospital

Stony Brook, NY

- Built a hospital website using **React/Flask**, with **APIs, SQL/NoSQL data, and authentication**.
- Collaborated with BS/MD and pre-med students to analyze **de-identified patient records**, curated **datasets**, and **trained/evaluated clinical AI models**, incorporating feedback to improve **model quality**.
- Wrote clear docs, helped teammates and staff use the tools, watched **error logs**, and fixed issues to keep everything **stable**.

Secure Distributed Computation and Learning Networks Research

August 2024 – Present

Undergraduate Researcher

Stony Brook, NY

- Maintained **80% consensus** in **Byzantine networks** by projecting updates onto **convex sets** and eliminating **adversarial values**
- Achieved **77% valid input preservation** in **real-time decision-making** under **adversarial corruption**
- Enabled **stable updates** with **93% agent interaction** through **controlled neighborhood protocols**

RESEARCH PAPERS

NFLPA Analytics – Special Teams Workload and Injury Risk

October, 2025 - January, 2026

Python, DuckDB, Statsmodels, Poisson and Negative Binomial GLMs, Fixed Effects

- Analyzed special teams workload and next-week injury risk using **5,950 NFL team-weeks**, estimating fixed-effects Poisson and Negative Binomial models with clustered and bootstrap-validated inference.
- Shock workload weeks raised next-week injury probability by **2.52 pp (offense)** and **1.39 pp (defense)**, while workload volatility (25th–75th percentile) increased offensive injury risk by **2.46 pp** with no meaningful defensive effect.
- League-level scaling implies **+8.8 offensive** and **+5.7 defensive injuries per season** attributable to shock workloads; results documented in a research paper submitted to the NFL Players Association.

PROJECTS

RealityRPG - AI Gamified Goal-Tracking Platform (Hackathon Winner)

February 2026

React, Next.js, Three.js, Gaussian Splatting, Luma AI, Gemini, LangChain

- Enabled immersive goal tracking by building an AI-powered platform that converts personal goals into **interactive 3D environments** using **Gaussian Splatting reconstruction** to visually represent user progress and milestones.
- Generated navigable 3D worlds by developing a **reconstruction pipeline** that processes user-uploaded images with **Luma AI** and renders **real-time scenes in Three.js** for browser-based exploration.
- Integrated **Gemini agents** to generate dynamic **quests, feedback, and progress tracking** from user goals.

SurveiLens - Real-Time AI Safety Platform (Hackathon Winner)

November 2025

React, Flask, NeuralSeek, Snowflake, ElevenLabs, OpenCV, Gemini, VADER, WebRTC/RTSP, Vultr

- Designed and implemented a **real-time agentic and multimodal AI system** that fuses **video, audio, and language signals** from live streams to generate **continuous risk assessments** for safety monitoring.
- Engineered **inference pipelines** using **OpenCV** for video analysis, **Gemini-based reasoning** for contextual interpretation, and **sentiment analysis on audio transcripts** to improve detection accuracy and reduce **cross-modal false positives**.
- Deployed scalable **RTSP/WebRTC stream ingestion** with **Snowflake-backed event storage** and **NeuralSeek-orchestrated alerting** to enable **low-latency monitoring** and automated risk notifications.